**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**
**UNIVERSITY OF BARISHAL**
**FINAL EXAMINATION-2020**
**Course Title: Artificial Intelligence**
**Course Code: CSE-3205**
3rd Year 1st Semester
Session: 2017-18

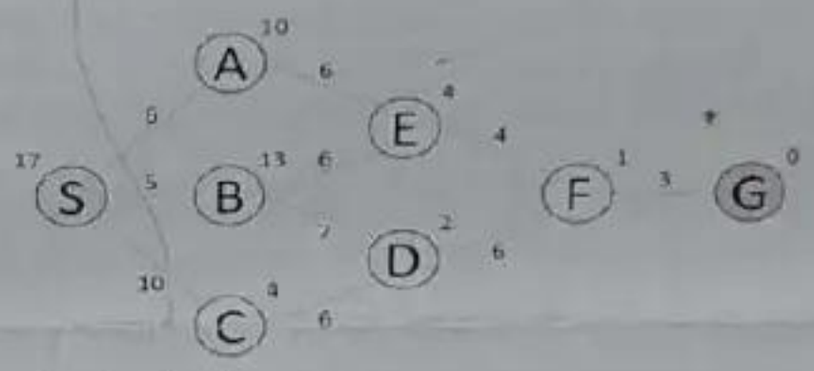
**Time:** 3 hours **Marks:** 60

**Answer any five Questions from the followings.**

1. a) The Task Environment for a **Virtual English Tutor** must have Performance, Environment, Actuators, and Sensors. By considering these four parameters develop the **PEAS** descriptions for a **Virtual English Tutor** Agent. [3]
   b) Define Agent and Rational Agent through real-time example [3]
   c) Distinguish between the following properties of a task environment: [3]
      i. Static or dynamic ii. Discrete or continuous iii. Single or multi-agent
   d) Briefly explain the PEAS description of the task environment for **an automated taxi.** [3]

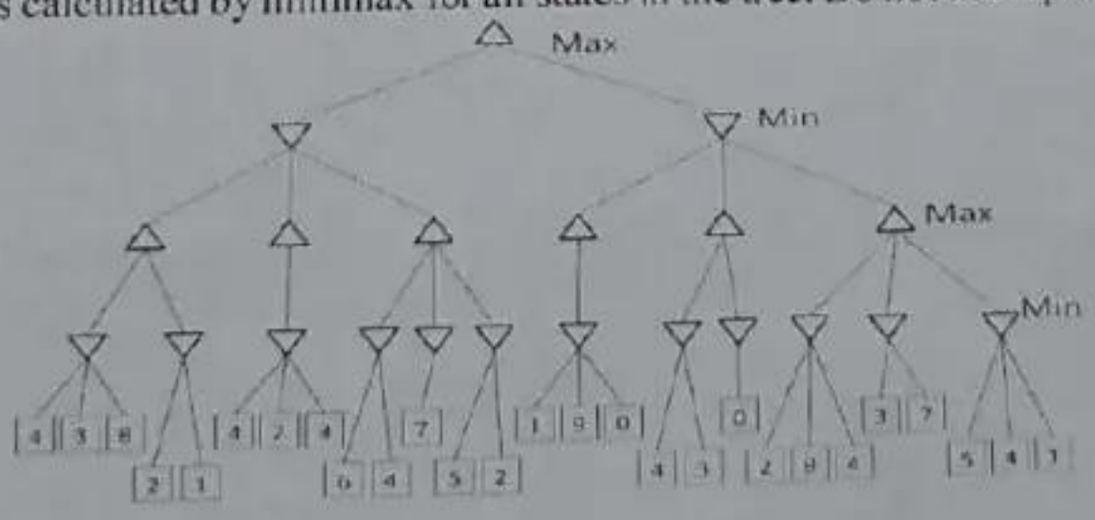
2. a) List the criteria to measure the performance of different search strategies. [3]
   b) Differentiate Uninformed Search (Blind search) and Informed Search (Heuristic Search) strategies. [3]
   c) Prove that the time and space complexity of BFS is $O(b^d)$. And explain it with an example [4]
   d) Does artificial intelligence provide an alternative to the biological substances? [2]

3. a) Prove that A* search is optimal if h(n) is admissible. [3]
   b) Perform A* Algorithm on the following figure. Explicitly write down the queue at each step [5]




   c) A hypothetical state space with heuristic evaluation is given as: [4]
   ROOT: A-5
   LEVEL 1: B-4, C-4, D-6
   LEVEL2: E-5, F-5, G-4, H-3, I, J
   LEVEL3: K, L, M, N, O-2, P-3, Q-3, R
   LEVEL4: S, T, U, V
   If P is the goal state, how do you reach the goal state using best-first search?

4. a) Give the values calculated by minimax for all states in the tree. Do not use alpha-beta pruning [5]




   b) Indicate which branches of the tree will be pruned by alpha-beta pruning? [4]
   c) Abduction is an inference rule that infers P from P→Q and Q. show that abduction is not sound. [3]

5. a) You are planning to purchase motorcycle. You look for a particular mode from three dealers. The probabilities of visiting a dealer and making a purchase are given in the following table. What is the probability that you buy the motorcycle from dealer D3? [4]

| Dealer | Visiting (%) | Purchase (%) |
|---|---|---|
| D1 | 0.2 | 0.4 |
| D2 | 0.3 | 0.4 |
| D3 | 0.5 | 0.2 |

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